

Optimal Execution Strategies Incorporating Internal Liquidity Through Market Making

Abstract

This paper proposes a novel algorithmic execution model that integrates interbank limit and market orders with internal liquidity provided through market making. Building upon a widely used framework for optimal execution, the model incorporates market impact for interbank orders while excluding it for internal market-making executions. The objective is to optimize the allocation between interbank and internal liquidity, thereby reducing market impact and enhancing execution efficiency.

JEL Classification: C41, G11 Mathematical Subject Classification (2010): 65C05, 60G40 Keywords: Algorithmic Trading, Market Impact, Optimal Execution, Liquidation, Stochastic Control, Impulse Control

1 Introduction

When executing large orders, carrying out trades over a short period can trigger unfavorable market movements, known as market impact. On the other hand, spreading the execution over a longer time exposes the trader to market risk. Algorithmic execution strategies must navigate this trade-off, and such strategies are generally categorized into static strategies, which follow a predetermined execution schedule, and dynamic strategies, which adjust the schedule in response to market conditions.

The seminal work of Almgren and Chriss [1] and Bertsimas and Lo [3] introduced a foundational model in this field, representing the market mid-price as a continuous diffusion process with market impact incorporated as a drift term. This model provides a framework for deriving static optimal execution strategies using variational methods. Regarding static strategies, many studies have also extended the modeling of market impact structures, including nonlinear market impact and resilience and transient impact in Dang[8], Gatheral et al. [9], Gatheral and Schied [10], and Curato et al.[7].

In real markets, it is critical to account for bid-offer spreads and to optimally balance the use of market orders(MOs) and limit orders(LOs). Cartea et al.[6], Cartea and Jaimungal [5] extended this framework by modeling LOs using a jump-diffusion process incorporating Poisson arrivals, while representing MOs as interventions within an impulse control framework. They solved the associated Hamilton-Jacobi-Bellman Quasi-Variational Inequality (HJB QVI) to derive dynamic strategies.

In financial institutions, market making(MM) plays an important role in providing liquidity to the market by simultaneously quoting buy and sell orders and acting as a

counterparty to other participants. Algorithmic execution can leverage internal liquidity through market making, attracting order flow by offering more favorable prices. This approach, particularly in OTC markets such as the FX market, reduces market impact and enables efficient execution compared to trading directly in external markets. Moreover, this mechanism benefits both the institution’s algorithmic execution clients and the clients receiving quotes on the market-making side. As such, it has become a widely adopted practice in the industry.

There has been extensive research on market making in Avellaneda and Stoikov[2], Guéant et al.[11] and Cartea et al.[4]. All of which focuses on determining optimal strategies for market makers. This paper, however, examines optimal strategies from the perspective of an algorithmic execution agent, rather than considering the profitability of market makers. Since this method essentially transfers the flow to algorithmic execution clients, it does not generate profit and loss (PL) for market makers.

This paper proposes a novel model that incorporates not only interbank MOs and LOs but also internal liquidity generated through MM as potential execution options. The motivation for this approach lies in the ability to quantify the optimal extent of favorable pricing in market making and the ideal integration of interbank and internal liquidity during execution. Our model builds upon the Cartea et al.[6] framework by extending it to account for market impact in interbank LOs and MOs, while excluding market impact for internal orders executed through MMs. This distinction allows us to analyze the interaction between interbank and internal liquidity. We derive the corresponding HJB QVI, solve it numerically, and analyze the resulting strategies to shed light on their characteristics.

2 Utilization of liquidity in the market making

This chapter provides a detailed explanation of algorithmic execution utilizing market making. Financial institutions operate both as market makers and takers. Specifically, as market makers, they primarily supply liquidity to client markets, and the resulting positions are hedged in the interbank market as market takers. Additionally, algorithmic execution services provided by financial institutions typically involve acting as a taker to execute large client orders on their behalf.

Such algorithmic execution is generally carried out through limit orders (LO) or market orders (MO) in the interbank market. If the institution is also engaged in market making, it can attract necessary flow by offering competitive prices to its market-making clients. This flow can then be handed over to the algorithmic execution side, creating mutually beneficial transactions for both the clients using algorithmic execution services and the clients engaging with the institution’s market-making services. This integrated approach enhances efficiency and value for all parties involved.

Figure 1 shows an illustration of algorithmic execution utilizing market making.

- FX market image (OTC), USD/JPY.
- Client order: Sell algorithmic execution.
- Agent of algorithmic execution requests internal market maker to provide favorable price to the client of market making side.

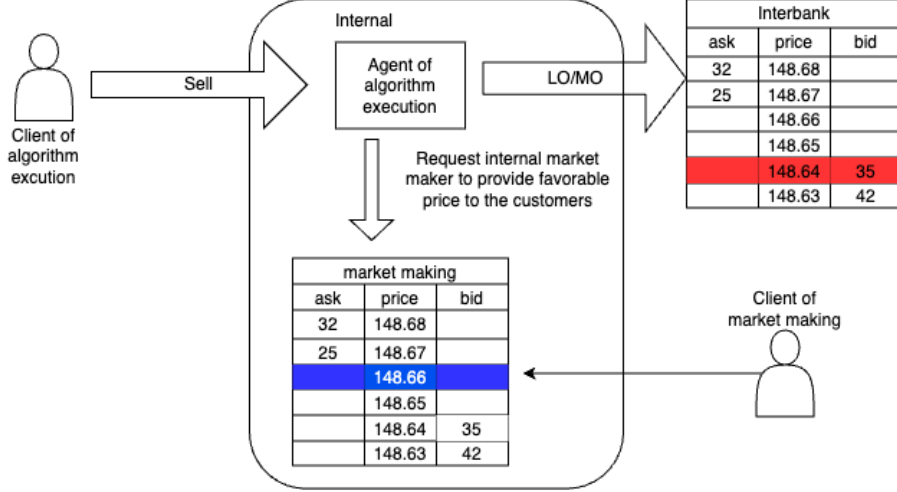


Figure 1: Image of algorithm execution utilizing MM

- For clients on the maker side, they can purchase at a special discounted price, which is lower than the usual ask price.
- For clients utilizing algorithmic execution, they can sell at a price higher than the interbank bid price while also minimizing market impact.

3 Model

3.1 Mathematical Framework: Combined Stochastic Control and Impulse Control

The proposed execution problem combines continuous stochastic controls, corresponding to the choice of spreads for limit and internal market-making orders, together with impulse controls representing market-order executions. Such problems naturally fall into the framework of combined stochastic and impulse control problems studied in Øksendal and Sulem [12] and Seydel [13]. In particular, the associated value function is characterized through a Hamilton–Jacobi–Bellman quasi-variational inequality (HJB-QVI) in the viscosity solution framework.

In this section, we summarize the standard framework for combined stochastic and impulse control problems developed in Seydel [13], which provides the theoretical foundation for the HJB-QVI formulation used in this paper.

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, P)$ be a filtered probability space satisfying the usual conditions. Let A be a compact non-empty metric space. The stochastic control process

$$v : [0, T] \times \Omega \rightarrow A$$

is assumed to be $\{\mathcal{F}_t\}$ -predictable. Let W be a d_1 -dimensional $\{\mathcal{F}_t\}$ -Brownian motion. Let N be a d_2 -dimensional counting process with compensator

$$\int_0^t \lambda(v_u) du,$$

where

$$\lambda : A \rightarrow \mathbb{R}_{\geq 0}^{d_2}$$

is a continuous function. More precisely,

$$N_t - \int_0^t \lambda(v_u) du$$

is assumed to be a $\{\mathcal{F}_t\}$ -martingale.

The impulse control is represented by a sequence

$$\theta = \{(\tau_i, \zeta_i)\}_{i \geq 1},$$

where τ_i are $\{\mathcal{F}_t\}$ -stopping times and ζ_i are $\mathcal{F}_{\tau_i}$ -measurable intervention variables taking values in a compact intervention set $\mathcal{Z}$.

The k -dimensional state process Y follows the stochastic differential equation with impulses

$$dY_t = \mu(Y_{t-}, v_{t-})dt + \sigma(Y_{t-}, v_{t-})dW_t + \nu(Y_{t-}, v_{t-})dN_t, t \in (\tau_{i-1}, \tau_i), \quad (1)$$

$$Y(\tau_i) = \Gamma(\check{Y}(\tau_i-), \zeta_i), i = 1, 2, \dots, \quad (2)$$

where the term $\check{Y}_{\tau_i-}$ denotes the value of the controlled Y in τ_i with a possible jump of the stochastic process, but without the impulse, i.e., $Y_{\tau_i-} + \Delta Y_{\tau_i}$, $\Gamma : \mathbf{R}^k \times \mathcal{Z} \rightarrow \mathbf{R}^k$, $\mu : \mathbf{R}^k \times A \rightarrow \mathbf{R}^k$, $\sigma : \mathbf{R}^k \times A \rightarrow \mathbf{R}^{k \times d_1}$ and $\nu : \mathbf{R}^k \times A \rightarrow \mathbf{R}^{k \times d_2}$ satisfying the necessary conditions such that existence and uniqueness of the SDE (1) with an initial value $y \in \mathbf{R}^k$ is guaranteed.

Let $\mathcal{A}(s, y)$ denote the set of admissible combined stochastic and impulse controls (v, θ) for which the controlled state process (1) is well defined and existence and uniqueness are guaranteed. For a sufficiently smooth function ψ , define the intervention operator

$$\mathcal{M}\psi(y) = \sup_{\zeta \in \mathcal{Z}} \psi(\Gamma(y, \zeta)).$$

The infinitesimal generator associated with the controlled dynamics between interventions is given by

$$L^v \psi(y) = \mu(y, v) \partial_y \psi(y) + \frac{1}{2} \sigma(y, v)^2 \partial_{yy} \psi(y) + (\psi(y + \nu(y, v)) - \psi(y)) \lambda(t, y, v).$$

We consider the value function

$$H(t, y) = \sup_{(v, \theta) \in \mathcal{A}(t, y)} E \left[g(Y_T) + \int_t^T f(u, Y_u, v_u) du \middle| \mathcal{F}_t \right], \quad (3)$$

where

$$f : \mathbf{R}^+ \times \mathbf{R}^k \times A \rightarrow \mathbf{R}, \quad g : \mathbf{R}^k \rightarrow \mathbf{R}$$

are measurable reward functions.

According to the dynamic programming principle for combined stochastic and impulse control problems, we introduce the following formal HJB-QVI associated with the value function.

Definition 1 (Formal HJB-QVI). *The formal Hamilton–Jacobi–Bellman quasi-variational inequality associated with the value function (3)*

$$\max \left\{ \partial_t H(t, y) + \sup_{v \in A} (L^v H(t, y) + f(t, y, v)), \mathcal{M}H(t, y) - H(t, y) \right\} = 0, \quad (4)$$

with terminal condition

$$H(T, y) = g(y).$$

Following Seydel [13], we impose the following assumptions on the controlled dynamics and reward functions.

(A1) The stochastic control set A and intervention set $\mathcal{Z}$ are compact.

(A2) The coefficients

$$\mu, \sigma, \nu, \Gamma, \lambda$$

are continuous in the state and control variables and satisfy standard continuity and growth conditions ensuring existence and uniqueness of the controlled process.

(A3) The reward functions

$$f(t, y, v), \quad g(y)$$

are continuous and satisfy polynomial growth conditions.

(A4) The associated HJB-QVI admits a comparison principle within the corresponding growth class.

The following theorem summarizes Theorem 2.2 of Seydel [13].

Theorem 2 (Seydel [13]). *Suppose that assumptions (A1)–(A4) hold. Then the value function*

$$H(t, y) = \sup_{(v, \theta) \in \mathcal{A}(t, y)} E \left[g(Y_T) + \int_t^T f(u, Y_u, v_u) du \middle| \mathcal{F}_t \right]$$

is the unique viscosity solution of the HJB-QVI (4).

3.2 Proposed Execution Model with Internal Market-Making Liquidity

We extend the Limit and Market Order model in Cartea et al. [6]. The setting involves selling a fixed notional amount Q_0 of the underlying asset (a similar setup applies to buying).

- T is the terminal time at which the liquidation ends.
- $X = (X_t)_{0 \leq t \leq T}$ denotes the agent's cash process with $X_0 = 0$.
- $S = (S_t)_{0 \leq t \leq T}$ denotes the mid price process of the asset, where $S_0 > 0$.
- $Q = (Q_t)_{0 \leq t \leq T}$ denotes the inventory process representing the remaining inventory to be liquidated.
- $\delta^L = (\delta_t^L)_{0 \leq t \leq T}$ denotes the depth at which the agent posts limit sell orders. Namely, the agent posts limit orders at the price $S_t + \delta_t^L$ at time t .
- $\delta^I = (\delta_t^I)_{0 \leq t \leq T}$ denotes the ask spread shown through the internal market-making desk. Namely, the agent requests the market maker to provide the price $S_t + \delta_t^I$ at time t .
- The spread controls

$$\delta = (\delta^L, \delta^I)$$

are assumed to be $\{\mathcal{F}_t\}$ -predictable and to take values in the compact admissible set

$$D := [\underline{\delta}^L, \bar{\delta}^L] \times [\underline{\delta}^I, \bar{\delta}^I].$$

- $N^L = (N_t^L)_{0 \leq t \leq T}$ denotes the counting process associated with executions of the agent's limit orders.
- $N^I = (N_t^I)_{0 \leq t \leq T}$ denotes the counting process associated with internal executions through market making.
- Since the problem is formulated as a liquidation problem, we impose the inventory constraint

$$Q_t \geq 0.$$

Accordingly, executions through limit orders and internal market making are disabled once the inventory reaches zero.

- The counting processes are characterized through their compensators:

$$N_t^L - \int_0^t \lambda^L e^{-\kappa \delta_u^L} \mathbf{1}_{\{Q_{u-} > 0\}} du$$

and

$$N_t^I - \int_0^t \lambda^I e^{-\kappa \delta_u^I} \mathbf{1}_{\{Q_{u-} > 0\}} du$$

are assumed to be $\{\mathcal{F}_t\}$ -martingales.

Here, $\lambda^L > 0$, $\lambda^I > 0$, and $\kappa > 0$ are constants. The quantities

$$\lambda^L e^{-\kappa \delta_t^L}, \quad \lambda^I e^{-\kappa \delta_t^I}$$

can be interpreted as the execution rates of limit-order and internal executions, respectively. Larger values of δ_t^L and δ_t^I correspond to less aggressive quotes and therefore lower execution frequencies.

- $\theta = \{(\tau_i, \zeta_i)\}_{i \geq 1}$ denotes the impulse control corresponding to market-order executions. Here, τ_i represents the execution timing of the market order, and ζ_i is an $\mathcal{F}_{\tau_i}$ -measurable random variable representing the execution size.

We assume

$$\zeta_i \in \{0, 1, \dots, Q_{\tau_i-}\},$$

which guarantees that the inventory process remains non-negative. We formally set $\tau_0 = 0$.

- $\xi > 0$ denotes the bid spread, namely the distance between the mid price and the best bid price. For simplicity, ξ is assumed to be deterministic.
- $\alpha^L > 0$ denotes the market impact parameter associated with limit-order executions.
- $\alpha^M > 0$ denotes the market impact parameter associated with market-order executions.

The state process is given by

$$Y_t = (X_t, S_t, Q_t).$$

The controlled dynamics between impulse intervention times are

$$dX_t = (S_t + \delta_{t-}^L - \alpha^L \lambda_{t-}^L) dN_t^L + (S_t + \delta_{t-}^I) dN_t^I, \quad (5)$$

$$dS_t = \sigma dW_t, \quad (6)$$

$$dQ_t = -dN_t^L - dN_t^I, \quad (7)$$

for $t \in (\tau_{i-1}, \tau_i)$.

At impulse intervention times,

$$X_{\tau_i} = \tilde{X}_{\tau_i-} + (S_{\tau_i} - \xi)\zeta_i - \alpha^M \zeta_i^\beta, \quad (8)$$

$$S_{\tau_i} = S_{\tau_i-}, \quad (9)$$

$$Q_{\tau_i} = \tilde{Q}_{\tau_i-} - \zeta_i, \quad (10)$$

for $i = 1, 2, \dots$

Here,

$$\tilde{Y}_{\tau_i-} = Y_{\tau_i-} + \Delta Y_{\tau_i}$$

denotes the state immediately after the jump generated by the counting process but before the impulse intervention.

Equation (5) describes the increase in cash generated by executions through limit orders and internal market making. The term

$$-\alpha^L \lambda_t^L$$

represents the market impact associated with external limit-order executions. In contrast, no market impact is imposed on executions obtained through internal market making. This reflects the fact that internalization does not directly consume external market liquidity, which is particularly relevant in OTC markets such as FX markets.

Equation (6) assumes that the mid price follows a Brownian motion with volatility σ .

Equation (7) shows that the inventory decreases by one unit whenever either a limit-order execution or an internal execution occurs.

The impulse equations (8)–(10) represent market-order executions. At time τ_i , the agent executes a market order of size ζ_i at the bid price $S_{\tau_i} - \xi$, together with nonlinear market impact

$$\alpha^M \zeta_i^\beta.$$

Let $\mathcal{A}(s, y)$ denote the class of admissible controls (δ, θ) for which the controlled dynamics (5)–(7) admit a unique solution starting from the initial state

$$y = (x, s, q)$$

at time s .

The associated value function is defined by

$$H(t, x, s, q) = \sup_{(\delta, \theta) \in \mathcal{A}(t, y)} E \left[X_T + Q_T(S_T - \xi - \alpha Q_T) - \phi \int_t^T (Q_u - \bar{q}_u)^2 du \right]. \quad (11)$$

The objective functional in (11) consists of the following three components:

1. the terminal cash position X_T ;
2. the terminal liquidation value of the remaining inventory, where any remaining position is liquidated through market orders at terminal time together with terminal market impact costs;
3. the inventory-penalty term

$$\phi \int_t^T (Q_u - \bar{q}_u)^2 du,$$

which penalizes deviations from the benchmark inventory trajectory $\bar{q}$.

3.3 HJB-QVI Characterization of the Proposed Execution Model

We now verify that the proposed execution model introduced in Section 3.2 fits into the general framework summarized in Section 3.1. This verification is important because the compensators of the counting processes depend on the spread controls.

Proposition 3. *The proposed execution model introduced in Section 3.2 satisfies assumptions (A1)–(A4) in Section 3.1. Consequently, Theorem 2 applies.*

Proof. We verify assumptions (A1)–(A4) in Section 3.1.

(A1) The admissible spread-control set

$$D = [\underline{\delta}^L, \bar{\delta}^L] \times [\underline{\delta}^I, \bar{\delta}^I]$$

is compact by assumption. Moreover, for each inventory level q , the intervention set

$$\mathcal{Z}(q) = \{0, 1, \dots, q\}$$

is finite and therefore compact.

(A2) The coefficients of the controlled dynamics are continuous in the state and control variables. Since

$$0 \leq Q_t \leq Q_0,$$

the inventory process is bounded, and therefore all coefficients satisfy at most polynomial growth conditions in the remaining state variables (x, s) .

Furthermore, the counting processes are characterized through the compensators

$$\int_0^t \lambda^L e^{-\kappa \delta_u^L} \mathbf{1}_{\{Q_{u-} > 0\}} du,$$

and

$$\int_0^t \lambda^I e^{-\kappa \delta_u^I} \mathbf{1}_{\{Q_{u-} > 0\}} du,$$

which define the corresponding compensated counting processes as martingales. Hence the controlled dynamics are well defined.

(A3) The running reward and terminal reward functions are

$$f(t, x, s, q, \delta) = -\phi(q - \bar{q}_t)^2,$$

and

$$g(x, s, q) = x + q(s - \xi - \alpha q).$$

These functions are continuous and satisfy polynomial growth conditions.

(A4) The compensator coefficients

$$\lambda^L e^{-\kappa \delta_t^L}, \quad \lambda^I e^{-\kappa \delta_t^I}$$

are bounded on the compact control set. Moreover, the coefficients are locally Lipschitz continuous in the state variables. Therefore the regularity assumptions required for the comparison principle are satisfied. Hence assumptions (A1)–(A4) hold. $\blacksquare$

We now derive the HJB-QVI associated with the proposed execution problem.

Theorem 4. Assume that the conditions of Proposition 3 hold. Then the value function (11) associated with the proposed execution problem is the unique viscosity solution of the following HJB-QVI:

$$\begin{aligned} & \max \left\{ \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \frac{\partial^2 H}{\partial s^2} - \phi(q - \bar{q}_t)^2 \right. \\ & \quad + \sup_{\delta^L, \delta^I} \left([H(t, x + s + \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L}, s, q - 1) - H(t, x, s, q)] \lambda^L e^{-\kappa \delta^L} \right. \\ & \quad \quad \left. + [H(t, x + s + \delta^I, s, q - 1) - H(t, x, s, q)] \lambda^I e^{-\kappa \delta^I} \right) \\ & \quad \left. + \max_{\zeta \in \{0, 1, \dots, q\}} \left(H(t, x + (s - \xi) \zeta - \alpha^M \zeta^\beta, s, q - \zeta) - H(t, x, s, q) \right) \right\} = 0, \quad (12) \end{aligned}$$

subject to the boundary and terminal conditions

$$\begin{aligned} H(t, x, s, 0) &= x - \phi \int_t^T \bar{q}_u^2 du, \\ H(T, x, s, q) &= x + q(s - \xi - \alpha q). \end{aligned}$$

Proof. By Proposition 3, the proposed execution model satisfies assumptions (A1)–(A4) in Section 3.1. Hence Theorem 2 is applicable. For the dynamics (5)–(7), the continuous martingale part comes only from the mid-price process (6). Therefore the diffusion contribution to the infinitesimal generator is

$$\frac{1}{2} \sigma^2 \frac{\partial^2 H}{\partial s^2}.$$

A limit-order execution changes the state from (x, s, q) to

$$(x + s + \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L}, s, q - 1),$$

with compensator coefficient

$$\lambda^L e^{-\kappa \delta^L}.$$

Similarly, an internal execution changes the state from (x, s, q) to

$$(x + s + \delta^I, s, q - 1),$$

with compensator coefficient

$$\lambda^I e^{-\kappa \delta^I}.$$

Hence the infinitesimal generator is

$$\begin{aligned} L^\delta H &= \frac{1}{2} \sigma^2 \frac{\partial^2 H}{\partial s^2} \\ & \quad + [H(t, x + s + \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L}, s, q - 1) - H(t, x, s, q)] \lambda^L e^{-\kappa \delta^L} \\ & \quad + [H(t, x + s + \delta^I, s, q - 1) - H(t, x, s, q)] \lambda^I e^{-\kappa \delta^I}. \end{aligned}$$

The impulse intervention corresponding to a market order of size ζ maps the state (x, s, q) to

$$(x + (s - \xi)\zeta - \alpha^M \zeta^\beta, s, q - \zeta),$$

where $\zeta \in \{0, 1, \dots, q\}$. Therefore the intervention operator is

$$\mathcal{M}H(t, x, s, q) = \max_{\zeta \in \{0, 1, \dots, q\}} H(t, x + (s - \xi)\zeta - \alpha^M \zeta^\beta, s, q - \zeta).$$

The running reward and terminal reward functions are

$$f(t, x, s, q, \delta) = -\phi(q - \bar{q}_t)^2, \quad g(x, s, q) = x + q(s - \xi - \alpha q).$$

Substituting these expressions into the general HJB-QVI (4) yields (12) together with the stated boundary and terminal conditions. Finally, Theorem 2 implies that the value function is the unique viscosity solution of (12). $\blacksquare$

3.4 Reduction of the HJB-QVI and Construction of the Optimal Strategy

Following Cartea et al. [6], we seek a solution of the HJB-QVI (12) through an ansatz consistent with the affine structure of the terminal and boundary conditions. More precisely, motivated by

$$H(T, x, s, q) = x + q(s - \xi - \alpha q),$$

and

$$H(t, x, s, 0) = x - \phi \int_t^T \bar{q}_u^2 du,$$

we assume that the value function admits the representation

$$H(t, x, s, q) = x + qs + h(t, q),$$

where the function $h(t, q)$ represents the nonlinear inventory-dependent correction term. This ansatz separates the linear dependence on the cash position x and the mid price s from the genuinely nonlinear inventory optimization component. Consequently, the original four-dimensional HJB-QVI can be reduced to a lower-dimensional equation for $h(t, q)$.

Proposition 5. *Suppose that the value function admits the representation*

$$H(t, x, s, q) = x + qs + h(t, q).$$

Then the HJB-QVI (12) reduces to

$$\begin{aligned} & \max \left\{ \partial_t h(t, q) - \phi(q - \bar{q}_t)^2 \right. \\ & \quad + \sup_{\delta^I} \left((\delta^I + h(t, q - 1) - h(t, q)) \lambda^I e^{-\kappa \delta^I} \right) \\ & \quad + \sup_{\delta^L} \left((\delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L} + h(t, q - 1) - h(t, q)) \lambda^L e^{-\kappa \delta^L} \right) \\ & \quad \left. + \max_{\zeta \in \{0, 1, \dots, q\}} \left(h(t, q - \zeta) - h(t, q) - (\xi \zeta + \alpha^M \zeta^\beta) \right) \right\} = 0, \end{aligned} \quad (13)$$

with boundary and terminal conditions

$$\begin{aligned} h(t, 0) &= -\phi \int_t^T \bar{q}_u^2 du, \\ h(T, q) &= -q(\xi + \alpha q). \end{aligned}$$

Proof. We first observe that the optimization over (δ^L, δ^I) can be separated as

$$\begin{aligned} & \sup_{\delta^L, \delta^I} \left([H(t, x + s + \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L}, s, q - 1) - H(t, x, s, q)] \lambda^L e^{-\kappa \delta^L} \right. \\ & \quad \left. + [H(t, x + s + \delta^I, s, q - 1) - H(t, x, s, q)] \lambda^I e^{-\kappa \delta^I} \right) \\ &= \sup_{\delta^L} \left([H(t, x + s + \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L}, s, q - 1) - H(t, x, s, q)] \lambda^L e^{-\kappa \delta^L} \right) \\ & \quad + \sup_{\delta^I} \left([H(t, x + s + \delta^I, s, q - 1) - H(t, x, s, q)] \lambda^I e^{-\kappa \delta^I} \right). \end{aligned}$$

We then substitute the ansatz

$$H(t, x, s, q) = x + qs + h(t, q)$$

into (12). Since

$$\frac{\partial H}{\partial t} = \partial_t h(t, q), \quad \frac{\partial^2 H}{\partial s^2} = 0,$$

the diffusion term disappears. Moreover,

$$\begin{aligned} & H(t, x + s + \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L}, s, q - 1) - H(t, x, s, q) \\ &= \delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L} + h(t, q - 1) - h(t, q), \end{aligned}$$

and similarly

$$\begin{aligned} & H(t, x + s + \delta^I, s, q - 1) - H(t, x, s, q) \\ &= \delta^I + h(t, q - 1) - h(t, q). \end{aligned}$$

For the impulse intervention term,

$$\begin{aligned} & H(t, x + (s - \xi)\zeta - \alpha^M \zeta^\beta, s, q - \zeta) - H(t, x, s, q) \\ &= h(t, q - \zeta) - h(t, q) - (\xi\zeta + \alpha^M \zeta^\beta). \end{aligned}$$

Substituting these expressions into (12) yields (13). The boundary and terminal conditions follow directly from those of (12). $\blacksquare$

The reduced equation (13) allows us to characterize the optimal controls explicitly or semi-explicitly.

Proposition 6. *Suppose that $h(t, q)$ solves (13). Then the optimal internal market-making spread δ_t^{I*} is characterized by*

$$\frac{\partial}{\partial \delta^I} \left((\delta^I + h(t, q - 1) - h(t, q)) \lambda^I e^{-\kappa \delta^I} \right) = 0,$$

and is explicitly given by

$$\delta_t^{I*} = \frac{1}{\kappa} + h(t, q) - h(t, q - 1). \quad (14)$$

The optimal limit-order depth δ_t^{L*} is characterized as the solution of

$$\frac{\partial}{\partial \delta^L} \left((\delta^L - \alpha^L \lambda^L e^{-\kappa \delta^L} + h(t, q - 1) - h(t, q)) \lambda^L e^{-\kappa \delta^L} \right) = 0. \quad (15)$$

Furthermore, the optimal market-order size ζ^* satisfies

$$\zeta^* \in \arg \max_{\zeta \in \{0, 1, \dots, q\}} \left(h(t, q - \zeta) - h(t, q) - (\xi \zeta + \alpha^M \zeta^\beta) \right),$$

and the corresponding intervention timing is given by

$$\tau_i = \inf \{ t > \tau_{i-1}; h(t, q - \zeta^*) - h(t, q) = \xi \zeta^* + \alpha^M (\zeta^*)^\beta \}. \quad (16)$$

Proof. The characterization of the optimal controls follows from the first-order optimality conditions associated with the supremum terms in (13). Differentiating

$$(\delta^I + h(t, q - 1) - h(t, q)) \lambda^I e^{-\kappa \delta^I}$$

with respect to δ^I yields

$$\lambda^I e^{-\kappa \delta^I} (1 - \kappa (\delta^I + h(t, q - 1) - h(t, q))).$$

Setting this equal to zero gives (14). Similarly, the optimal limit-order depth δ_t^{L*} is obtained from the first-order condition (15). Finally, the optimal impulse intervention is determined by maximizing the impulse gain term in (13). The intervention timing corresponds to the first hitting time at which immediate intervention becomes optimal, yielding (16). $\blacksquare$

Based on these reductions, the original HJB-QVI problem is reduced to solving the lower-dimensional nonlinear equation (13) for the function $h(t, q)$. This equation can be solved numerically using standard numerical schemes for HJB equations and quasi-variational inequalities, for example finite-difference methods (FDM). Once the function $h(t, q)$ is obtained, the optimal execution strategy is fully characterized through

$$\delta_t^{I*}, \quad \delta_t^{L*}, \quad \tau_i.$$

In particular,

- δ_t^{I*} represents the optimal internal market-making spread;
- δ_t^{L*} represents the optimal depth of the limit orders;
- τ_i represents the optimal timing of market-order executions.

4 Numerical Experiments

We carry out a numerical implementation to compute the optimal execution strategy, including the execution timing of MO, the depth of LOs, and ask MM spread. We compare the following strategies.

1. ACAlmgren-Chriss model [1]. It is a static strategy $\bar{q}_t$,

$$\bar{q}_t = Q_0 \frac{\sinh(\gamma(T-t))}{\sinh(\gamma T)}. \quad (17)$$

2. LO/MO model by Cartea et al. [6], which includes benchmark (17) as a benchmark.
3. LO/MO/MM model proposed in section 3, which includes benchmark (17) as a benchmark in the equation (11).

Our implementation is found at the following link.

<https://gist.github.com/morimoto0606/20a199e41408455913ef91406b6d6861>

It is based on the sample code by Sebastian Jaimungal.

<https://sebastian.statistics.utoronto.ca/books/algo-and-hf-trading/code/>

The parameters are as follows.

$$T = 60, Q_0 = 10, \sigma = 0.01, \alpha = 0.0001, \phi = 0.001, \gamma = 0.1, \kappa = 100, \\ \lambda^L = 50/60, \lambda^I = 60/60, \alpha^L = 0.005, \alpha^M = 0.0001, \beta^M = 1.5, \xi = 0.01$$

Figure 2 shows the execution schedule by MO in the case where neither LO nor MM are filled for each strategy. The blue line represents Strategy 1 Almgren-Chriss benchmark, which corresponds to a static execution schedule used as a benchmark for Strategy 2 and 3. The red and green dots represent the times and inventories when MO are executed for Strategy 2 and 3, respectively. As for the execution methods, Strategy 2 uses both MO and LO, while Strategy 3 also incorporates internal orders, utilizing MM to further enhance profitability. As a result, Strategy 2 is behind the benchmark schedule of Strategy 1 due to the use of LO in addition to MO, and Strategy 3 is further delayed compared to Strategy 2, as it waits for internal orders through MM. Table 1 shows the actual time of red dot τ_L and green dot τ_I .

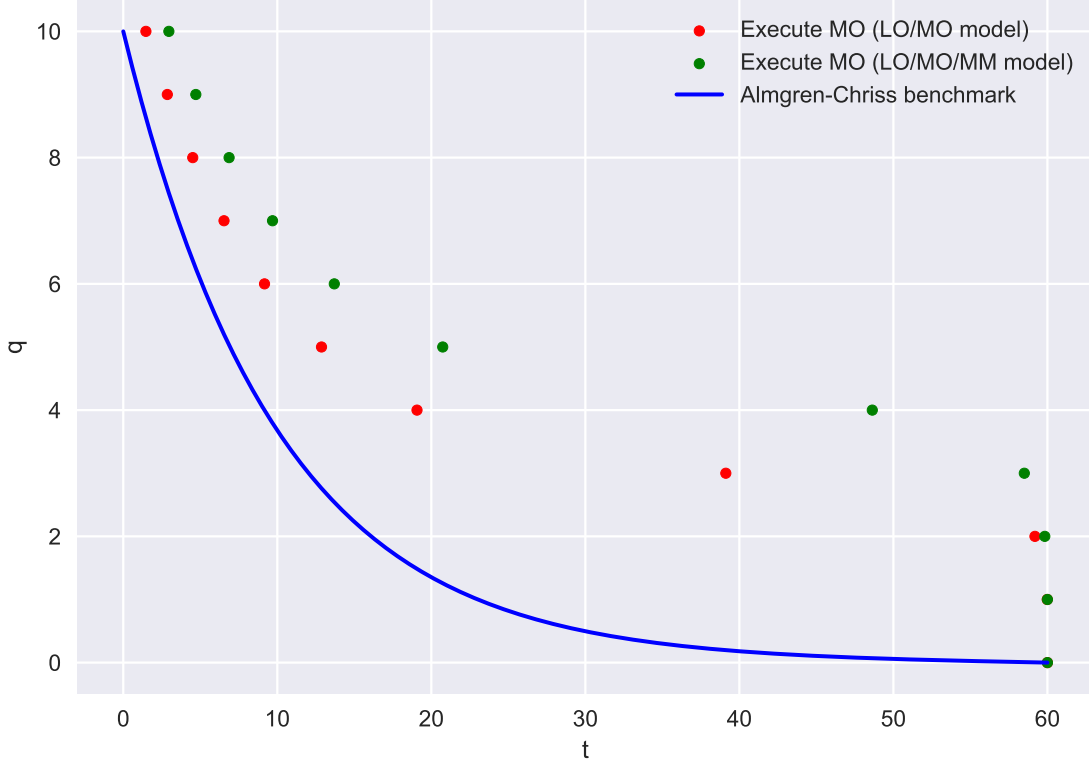


Figure 2: Optimal execution schedule and MO timing

Table 1: Optimal MO execution timing

q	10	9	8	7	6	5	4	3	2	1	0
$\tau_{\text{LO/MO}}$	1.47	2.86	4.51	6.54	9.17	12.87	19.07	39.12	59.19	59.99	60.00
$\tau_{\text{LO/MO/MM}}$	2.96	4.71	6.87	9.69	13.70	20.74	48.63	58.50	59.83	60.00	60.00

Table 2 shows the optimal market order execution size. Notice that in the LO/MO/MM model, the market order size ζ was also chosen optimally. ζ is a function of t and q , and here it represents the statistics for the sample of $q = 0$ to 10 at each $t = 10 - 50$. According to this, there is a trend that the execution size of the market order increases as time progresses. This can be attributed to the increasing need to complete the execution using the MO, even if the price is not as favorable as the LO or MM, due to the diminishing time remaining. Since market impact is incorporated as the 0.5 power of the execution size, it is considered more efficient to execute 2 units in one MO rather than executing 1 unit twice.”

Table 2: Statistics of Optimal market order size of LO/MO/MM model

Time	count	mean	std	min	25%	50%	75%	max
10	11.0	0.545455	0.687552	0.0	0.0	0.0	1.0	2.0
20	11.0	0.909091	1.044466	0.0	0.0	1.0	1.5	3.0
30	11.0	1.272727	1.348400	0.0	0.0	1.0	2.0	4.0
40	11.0	1.272727	1.348400	0.0	0.0	1.0	2.0	4.0
50	11.0	1.545455	1.368476	0.0	0.5	1.0	2.5	4.0

Table 3, 4 and Figure 3 shows, for Strategy 3, the optimal LO depth (dashed line) for each inventory at each time, as well as the optimal ask MM spread (solid line). For simplicity in presentation, only the curves for odd inventory levels are shown here. When comparing the LO depth δ^L and MM spread δ^I for the same inventory, they are nearly identical when there is still plenty of time remaining and the inventory is small around 1. On the other hand, when the remaining time is short and there is more inventory left, the MM spread is thinner than the LO depth (client-favorable) price, and it is particularly suggested that a better price can be offered, even beyond the mid. This suggests that there is no market impact during execution through MM, and since execution can occur at a better price than the MO, even beyond the mid, this behavior aligns well with real-world experience.

Table 3: LO depth

t	$\delta_t^{L*}(q=1)$	$\delta_t^L(q=3)$	$\delta_t^L(q=5)$	$\delta_t^L(q=7)$	$\delta_t^L(q=9)$
10	0.081970	0.029968	0.012390	0.004944	0.004944
20	0.044861	0.015076	0.005066	0.004944	0.004944
30	0.033752	0.011292	0.004944	0.004944	0.004822
40	0.029480	0.009949	0.004944	0.004944	0.004822
50	0.025574	0.008850	0.004944	0.004944	0.004822

Table 4: MM spread

t	$\delta_t^I(q=1)$	$\delta_t^I(q=3)$	$\delta_t^I(q=5)$	$\delta_t^I(q=7)$	$\delta_t^I(q=9)$
10	0.081992	0.029569	0.009957	-0.000100	-0.000100
20	0.044826	0.013250	0.000050	-0.000100	-0.000100
30	0.033527	0.008625	-0.00010	-0.000183	-0.000237
40	0.029097	0.006878	-0.00010	-0.000183	-0.000237
50	0.024879	0.005403	-0.00010	-0.000100	-0.000280

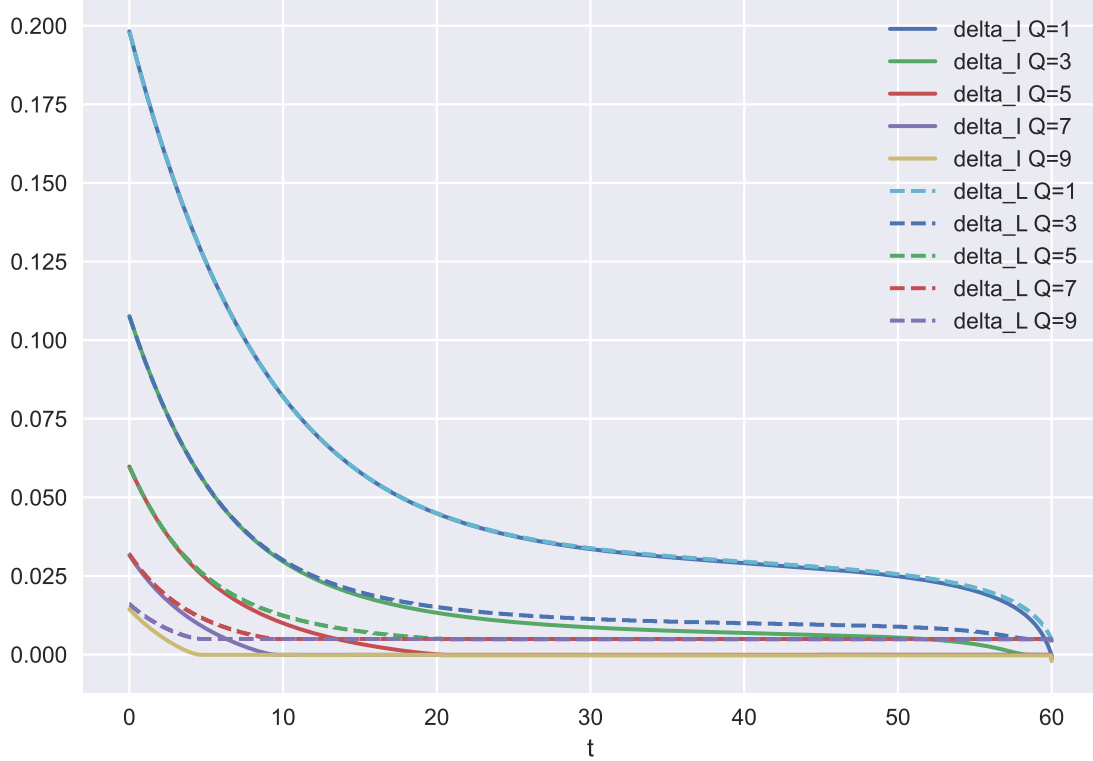


Figure 3: Optimal spread δ^I and depth δ^L

Table 5 reports the reduced value function $h(0, Q_0)$ obtained by solving the HJB-QVI backward in time. Recall that the value function is decomposed as

$$H(t, x, s, q) = x + qs + h(t, q),$$

where x denotes the cash position, qs is the mark-to-market value of the current inventory, and $h(t, q)$ represents the additional value generated by optimally executing the remaining inventory from time t onward.

In the comparison, we evaluate the value function at the normalized initial state $(x, s, q) = (0, S_0, Q_0)$. Since the cash component x enters the value function additively and the term $Q_0 S_0$ is common across all models, the difference between the models is entirely characterized by the reduced value function $h(0, Q_0)$. Therefore, comparing $h(0, Q_0)$ directly measures the difference in the optimal expected execution value generated by each execution mechanism.

In the numerical example, the reduced value is

$$h^{\text{LO/MO}}(0, Q_0) = -0.041139, \quad h^{\text{LO/MO/MM}}(0, Q_0) = 0.112355.$$

The LO/MO benchmark produces a negative reduced value, reflecting execution costs and inventory-risk penalties. In contrast, the inclusion of internal market-making liquidity turns the reduced value positive. Hence, incorporating internal market-making liquidity improves the reduced value by

$$h^{\text{LO/MO/MM}}(0, Q_0) - h^{\text{LO/MO}}(0, Q_0) = 0.153495.$$

This positive difference indicates that the additional internal execution channel increases the optimal value of the execution problem after accounting for market-order costs, limit-order execution opportunities, internal market-making executions, and inventory risk penalties.

Table 5: Comparison of reduced value functions at the initial inventory

Strategy	Reduced value $h(0, Q_0)$	Improvement over LO/MO
LO/MO	-0.041139	0.000000
LO/MO/MM	0.112355	0.153495

Table 6 and Figure 4 report sensitivity analyses of the reduced value function with respect to the internal execution intensity λ^I , the inventory penalty parameter ϕ , and the market-order impact parameter α^M .

First, the reduced value of the LO/MO/MM model increases monotonically as the internal execution intensity λ^I increases, as shown in Figure 4(a). This behavior is economically intuitive because a larger λ^I implies more frequent internal execution opportunities through market making, allowing the agent to complete liquidation with smaller market impact costs and reduced reliance on aggressive market orders. Consequently, the performance improvement over the benchmark LO/MO model also increases monotonically.

Second, as the inventory penalty parameter ϕ increases, the reduced values of both the LO/MO and LO/MO/MM models decrease, as shown in Figure 4(b). This is consistent with the interpretation of ϕ as a penalty for deviations from the benchmark liquidation trajectory. A larger inventory penalty reduces the incentive to wait for favorable executions through passive orders or internal liquidity, leading to more immediate liquidation behavior. As a result, the additional benefit obtained from internal market-making liquidity becomes smaller as ϕ increases.

Finally, Figure 4(c) shows that the reduced value exhibits only limited sensitivity with respect to the market-order impact parameter α^M within the considered parameter range. Although the reduced value decreases approximately linearly as α^M increases, the magnitude of the variation remains relatively small compared with the sensitivities to λ^I and ϕ . This suggests that, under the optimal strategy, market orders play a secondary role relative to limit-order and internal executions, and are mainly used as a supplementary execution mechanism near the terminal period.

Overall, the numerical results are consistent with the economic intuition underlying the proposed model and support the effectiveness of incorporating internal market-making liquidity into optimal execution strategies. In particular, the results indicate that the performance improvement is primarily driven by the availability of internal liquidity itself, rather than by the market-order mechanism.

Table 6: Sensitivity analysis of the reduced value function

Parameter	Value	$H^{\text{LO/MO}}(0)$	$H^{\text{LO/MO/MM}}(0)$	Improvement
λ^I	0.333	-0.041139	0.062264	0.103403
	0.667	-0.041139	0.090276	0.131415
	1.000	-0.041139	0.112355	0.153495
	1.333	-0.041139	0.130370	0.171509
	1.667	-0.041139	0.145538	0.186678
ϕ	0.00025	-0.072251	0.147240	0.219491
	0.00050	-0.056538	0.130043	0.186581
	0.00100	-0.041139	0.112355	0.153495
	0.00200	-0.023608	0.091985	0.115594
	0.00400	-0.000898	0.065818	0.066716
α^M	0.000025	-0.041139	0.112360	0.153500
	0.000050	-0.041139	0.112358	0.153498
	0.000100	-0.041139	0.112355	0.153495
	0.000200	-0.041139	0.112349	0.153488
	0.000400	-0.041139	0.112337	0.153477

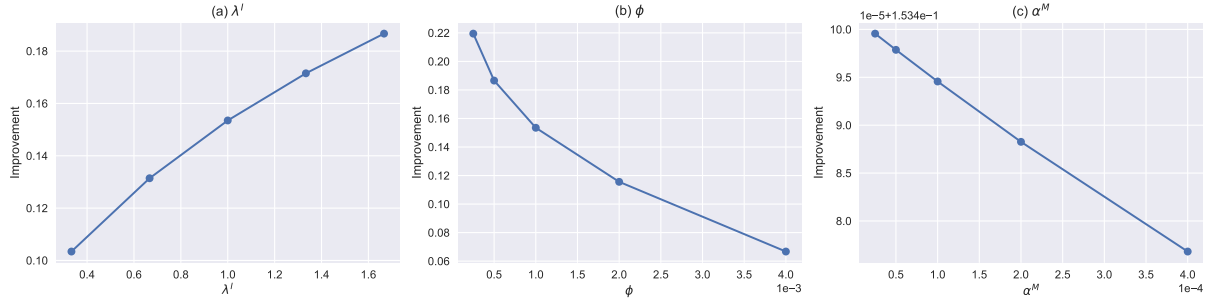


Figure 4: Sensitivity of the improvement over the LO/MO benchmark with respect to (a) internal execution intensity λ^I , (b) inventory penalty parameter ϕ , and (c) market-order impact parameter α^M .

References

- [1] Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–40, 2001.
- [2] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.
- [3] Dimitris Bertsimas and Andrew W Lo. Optimal control of execution costs. *Journal of financial markets*, 1(1):1–50, 1998.
- [4] Álvaro Cartea, Ryan Donnelly, and Sebastian Jaimungal. Algorithmic trading with model uncertainty. *SIAM Journal on Financial Mathematics*, 8(1):635–671, 2017.

- [5] Álvaro Cartea and Sebastian Jaimungal. Optimal execution with limit and market orders. *Quantitative Finance*, 15(8):1279–1291, 2015.
- [6] Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- [7] Gianbiagio Curato, Jim Gatheral, and Fabrizio Lillo. Optimal execution with non-linear transient market impact. *Quantitative Finance*, 17(1):41–54, 2017.
- [8] Ngoc-Minh Dang. Optimal execution with transient impact. *Market Microstructure and Liquidity*, 3(01):1750008, 2017.
- [9] Jim Gatheral, Alexander Schied, and Alla Slynko. Exponential resilience and decay of market impact. *Econophysics of Order-driven Markets: Proceedings of Econophysics-Kolkata V*, pages 225–236, 2011.
- [10] Jim Gatheral, Alexander Schied, and Alla Slynko. Transient linear price impact and fredholm integral equations. *Mathematical Finance: An International Journal of Mathematics, Statistics and Financial Economics*, 22(3):445–474, 2012.
- [11] Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: a solution to the market making problem. *Mathematics and financial economics*, 7:477–507, 2013.
- [12] Bernt Øksendal and Agnes Sulem. Stochastic control of jump diffusions. In *Applied Stochastic Control of Jump Diffusions*, pages 93–155. Springer, 2019.
- [13] Roland C Seydel. Existence and uniqueness of viscosity solutions for qvi associated with impulse control of jump-diffusions. *Stochastic Processes and their Applications*, 119(10):3719–3748, 2009.